ZARISKI'S MAIN THEOREM FOR SPACES: QUASI-FINITE SEPARATED MAPS FACTOR THROUGH A FINITE SPACE OVER QCQS TARGETS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7754-7768 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7770-7803, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete). The spaces ZMT card the derived-Chow and Stein machinery already lean on.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma quasi-finite-separated-quasi-affine; Limits of Spaces: integral-algebra-directed-colimit-finite, descend-opens, directed-inverse-system-has-limit, finite-type-eventually-closed; Morphisms of Spaces, lemma factor-the-other-way.

Source reference. Original lemma 02LR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7724–7738. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0827: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7803–7803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SF: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7803–7803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0828: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7803–7803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 082C: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7803–7803. Cited by the retained statement or

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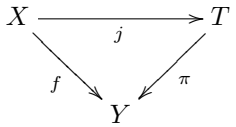
proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081U: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7803–7803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Zariski's Main Theorem)**.** *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-finite and separated and assume that Y is quasi-compact and quasi-separated. Then there exists a factorization*



where j is a quasi-compact open immersion and π is finite.

Proof. Let $X \to Y' \to Y$ be as in the conclusion of Lemma 02LR. By Limits of Spaces, Lemma 082C we can write $\nu_*\mathcal{O}_{Y'} = \operatorname{colim}_{i\in I} \mathcal{A}_i$ as a directed colimit of finite quasi-coherent $\mathcal{O}_X$-algebras $\mathcal{A}_i \subset \nu_*\mathcal{O}_{Y'}$. Then $\pi_i : T_i = \underline{\operatorname{Spec}}_Y(\mathcal{A}_i) \to Y$ is a finite morphism for each i. Note that the transition morphisms $T_{i'} \to T_i$ are affine and that $Y' = \lim T_i$.

By Limits of Spaces, Lemma 0827 there exists an i and a quasi-compact open $U_i \subset T_i$ whose inverse image in Y' equals $f'(X)$. For $i' \geq i$ let $U_{i'}$ be the inverse image of U_i in $T_{i'}$. Then $X \cong f'(X) = \lim_{i'\geq i} U_{i'}$, see Limits of Spaces, Lemma 07SF. By Limits of Spaces, Lemma 0828 we see that $X \to U_{i'}$ is a closed immersion for some $i' \geq i$. (In fact $X \cong U_{i'}$ for sufficiently large i' but we don't need this.) Hence $X \to T_{i'}$ is an immersion. By Morphisms of Spaces, Lemma 081U we can factor this as $X \to T \to T_{i'}$ where the first arrow is an open immersion and the second a closed immersion. Thus we win. □